Recipe Agent

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Roles

Scrum Master - TBD

Progress Reporting - Will be rotating per week

Github Link:

https://github.com/saroshawan100/Recipe-Agent

Project Scope

- An AI-powered chatbot that helps users decide what to cook based on the ingredients they already have.
- The user enters a list of available ingredients or an image of their fridge, and the Recipe Agent generates recipe suggestions using those ingredients.
- The user can specify a specific type of food, like vegetarian, italian, or soup.
- Generates a recipe based on input and creates an image of expected end result, as well as nutritional information like calories and macros.

Additional Features

Permanently stored information

- User Allergies
- User preferences
- Food dislikes
- Preferred cuisines
- Cooking skill level
- Nutrition goals

Relative stored information

- Available equipment
- Number of servings needed
- Available ingredients

Features

- Calories
- Macros
- Nutritional value
- Serving size
- Additional ingredients list for add ons (add chicken)

User Stories

- I want to have a nice warm cup of soup but only have what is in my fridge and do not want to go to the grocery store. What can I make that is low effort and hearty.
- I have milk that is going bad what can I make for lunch that uses three cups of milk and is a full meal.
- I have never made sushi before what do I need to get from the grocery store and how do I make sushi from scratch.
- My friend is coming over for dinner and is gluten free what can I make with what I have in my fridge (insert picture of fridge).

Task Allocation - Timeline

Phase 1 — Planning

- Define MVP
- Create user stories
- Design chatbot conversation flow
- Choose technology stack
- Set up GitHub/project repository
- Create outline slides

Phase 2 — Basic Agent

- Create chatbot interface
- Image ingredient input
- Connect AI model/API
- Create recipe bank for model learning
- Generate image of recipe
- Generate structured recipe responses

Phase 3 — Testing

- Generates usable recipes
- Uses the provided ingredients
- Doesn't hallucinate ingredients
- Gives understandable instructions
- Image recognition to help while user is cooking

Progress Update: Our First Working Build

What's working today, what's left,
and how to run it yourself

What We Built: Pantry Chef

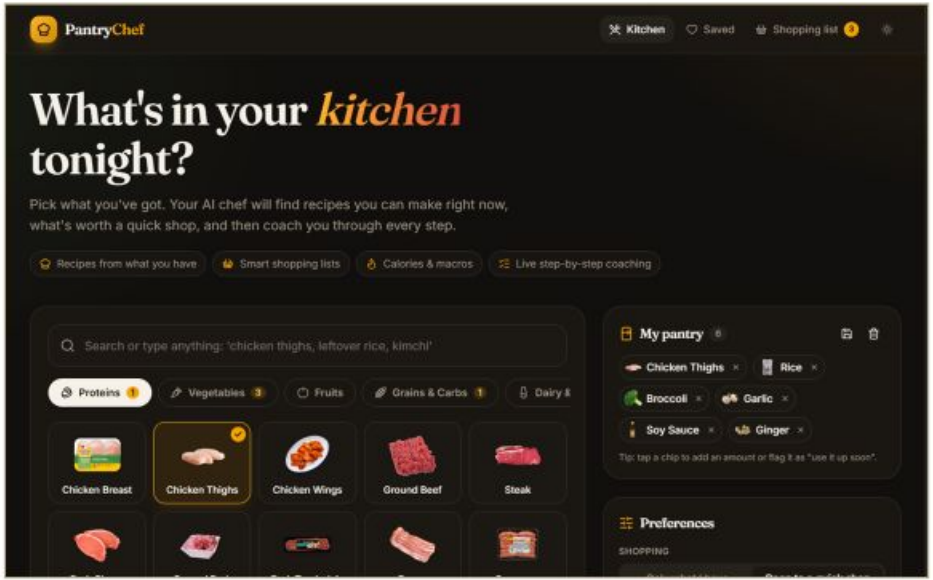
A working web app that turns your ingredients into recipes, then coaches you while you cook.

1. **Pick ingredients**
   Tap photos or type anything
2. **Set preferences**
   Diet, allergies, time, skill
3. **Get suggestions**
   Make it now, or worth a quick shop
4. **Open a recipe**
   Steps, nutrition, AI photo
5. **Cook with the Chef**
   Live, hands-free coaching

Also included: Ask the Chef chat on every page • Shopping list grouped by store aisle • Saved recipes

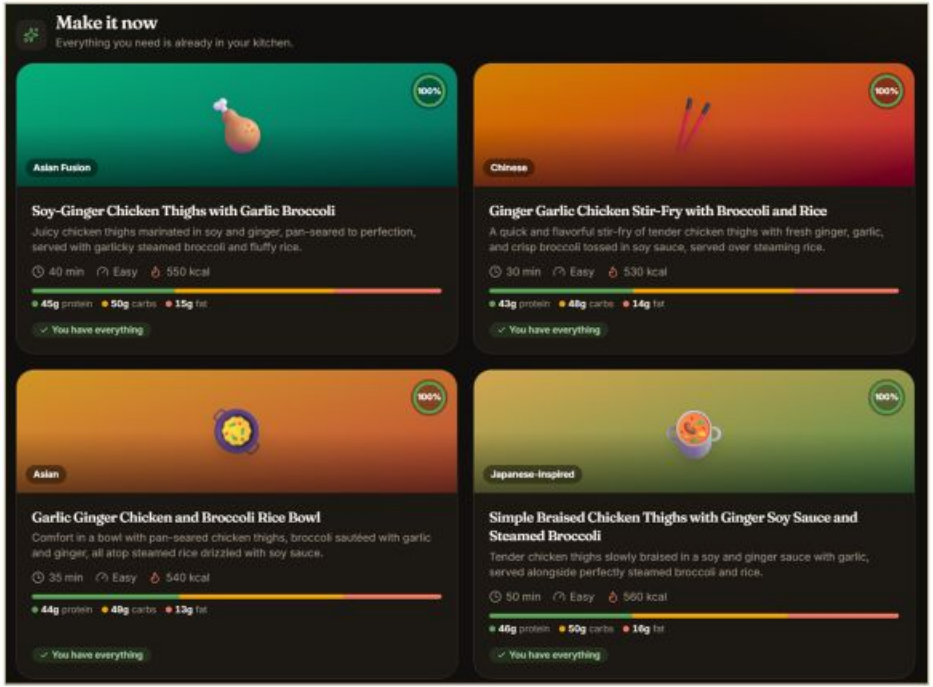
Step 1: Tell It What's in Your Kitchen

- **Pick from 195 ingredients** with photos in 8 categories, or type anything ("leftover rice, kimchi")
- **Add amounts or flag "use soon"** so recipes use those items first
- **Preferences:** diet, allergies, cuisine, equipment, servings, skill level, max cooking time, calorie and protein targets
- **Shopping mode:** "Only what I have" or "Open to a quick shop" with a limit on extra items
- **Free-text requests** like "spicy and cozy" or "meal-prep for the week"
- **Save your pantry** so it's there next time



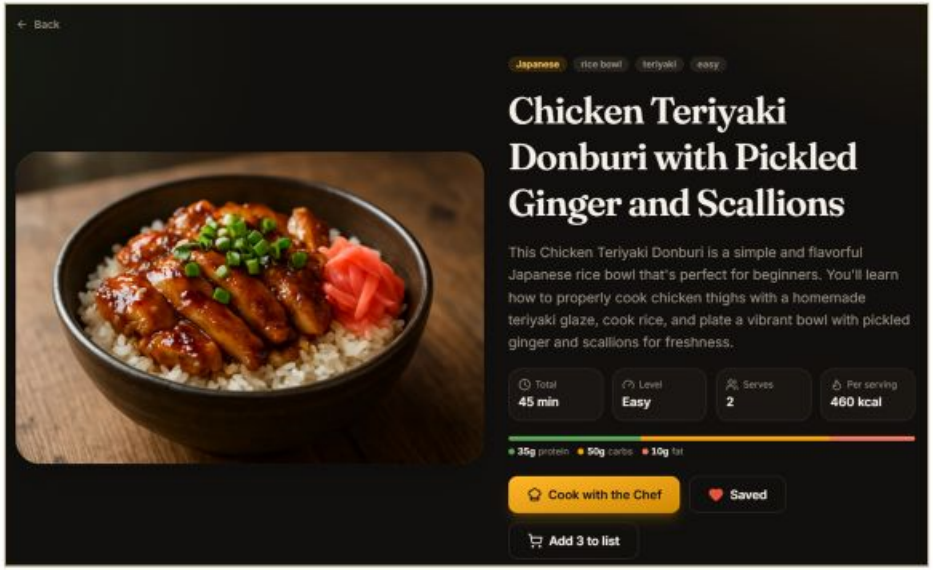
Step 2: Recipe Suggestions

- **Make it now:** recipes that need zero extra ingredients
- **Worth a quick shop:** recipes that need a few extra items, with exactly what to buy
- **Each card shows** how much of it you already have (match %), time, difficulty, calories, and protein, carbs and fat
- **A short chef's note** on what your pantry is great for
- **Respects diets and allergies,** and favors ingredients flagged "use soon"



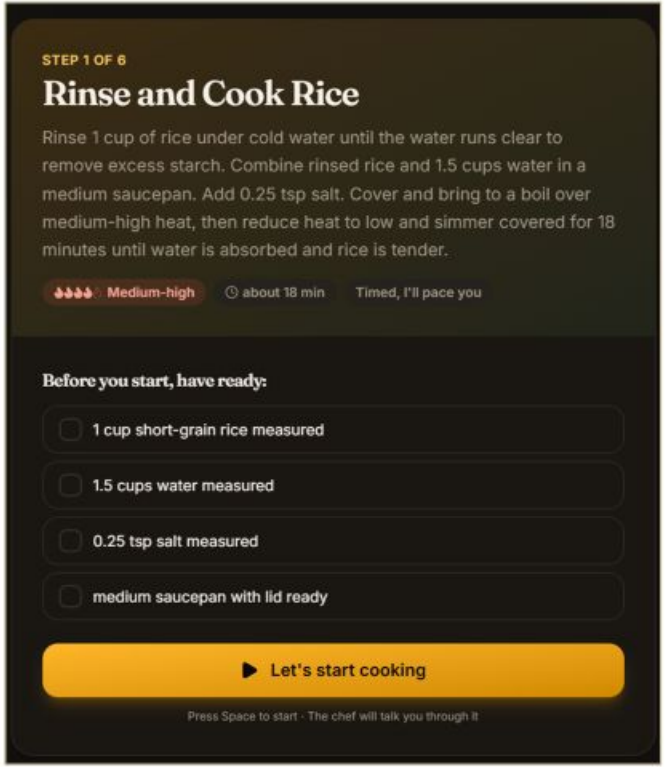
Step 3: The Full Recipe

- **AI-generated photo** of the finished dish
- **"You have" vs. "You need"** ingredient lists; add missing items to the shopping list in one tap
- **Beginner-friendly steps** with timers and doneness cues (what it should look, smell and sound like)
- **Nutrition per serving:** calories, macros, fiber, sugar and sodium
- **Glossary, substitutions, food-safety temperatures** and storage tips
- **Adjustable servings** rescale every ingredient amount



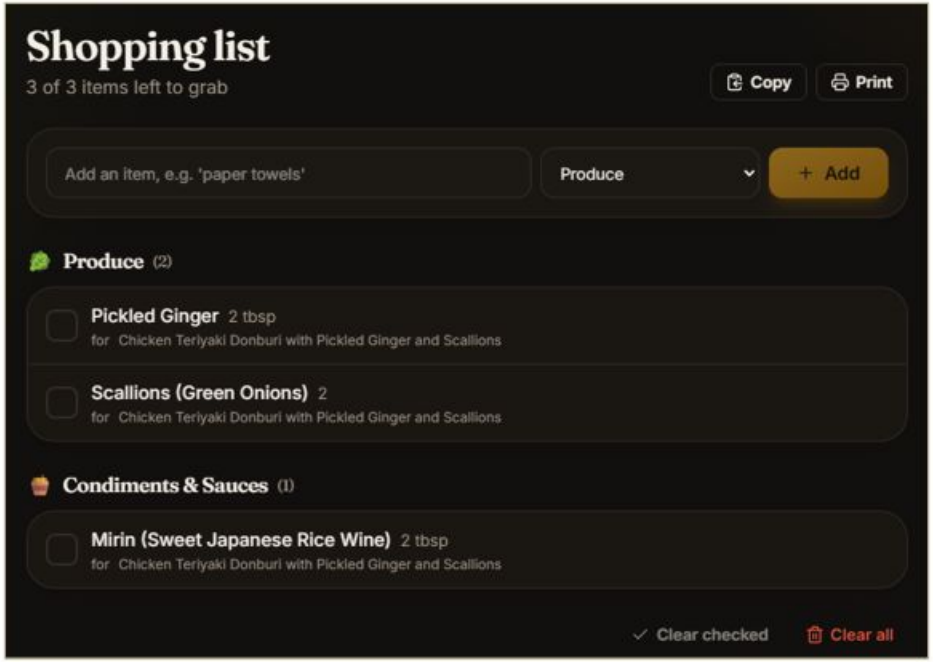
Step 4: Cook with the Chef

- **Live, hands-free coaching,** one step at a time
- **Minute-by-minute cues:** "flip the chicken now", "in 2:00 check for golden edges"
- **Doneness check-ins:** if it's not ready, the chef adds time and adjusts the plan
- **On-the-spot troubleshooting** ("the pan is smoking"), safety first
- **Speaks out loud,** plays chimes and keeps the screen awake
- **Prep checklist** before each step, and progress is saved



Shopping List & Ask the Chef

- **Shopping list** grouped by store aisle, showing which recipe each item is for
- **Add your own items,** check things off, then copy or print the list
- **Ask the Chef:** a streaming chat for substitutions, techniques, food safety, scaling and leftovers
- **Context-aware:** the chef knows your pantry and the recipe you're looking at
- **Saved recipes** are kept for later and never auto-deleted



How It's Built



- **Structured Outputs + Zod schemas:** the AI always returns well-formed recipes, nutrition and coaching
- **Caching:** recipe details, coaching scripts and photos are generated once, then reused
- **Private by design:** the OpenAI key stays on the server; no accounts, just an anonymous browser ID

Plan vs. Progress

✓ Done

- Recipes from ingredients on hand
- Food type filters (diet, cuisine, free text)
- AI image of the finished dish
- Calories, macros and nutrition
- Allergies, skill level, nutrition goals
- Equipment and servings
- Chatbot interface (Ask the Chef)
- Structured recipe responses
- GitHub repository set up

◑ Partly done

- Saved preferences: stored in the browser, no user accounts yet
- Food dislikes: free-text note only
- Add-ons ("add chicken"): covered by quick-shop recipes and substitutions
- No made-up ingredients: enforced in the AI instructions, still needs formal testing

○ Still to do

- Ingredient input from a photo of the fridge
- Recipe bank for model learning
- Image recognition while cooking
- Phase 3 testing

Proposed Next Steps

- **Phase 3 testing:** check that recipes are usable, use the given ingredients, don't invent ingredients, and have clear instructions
- **Fridge photo input:** let users upload a photo and detect ingredients automatically
- **Food dislikes:** add a dedicated preference instead of relying on free text
- **Accounts and syncing:** keep preferences and saved recipes across devices
- **Decide on the recipe bank:** keep it in scope, or rely on the AI model with structured outputs
- **Deploy:** host the app so the whole team and testers can use it without running it locally

Questions?

Thank you!